
intelligent job scheduling for hpc systems

Methodology and statistical framework. Honours
2026, Submission 2.

PRESENTER

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This deck reflects the state of the work at Submission 2, May 2026. It is restyled, not revised: the steps it describes in the future tense were still ahead at the time, and the results are in Submission 3 and the final paper.

recap: focus & approach ■

RECAP

RESEARCH FOCUS

identify the best-performing drl family for hpc scheduling across heterogeneous workloads

- Six DRL algorithms (PPO, DQN, A2C and their maskable variants)
- Two real Slurm traces (CPU and GPU workloads)
- Fixed allocator (best_fit) and time-aware splits

PROPOSED APPROACH

smoke → train → evaluate
→ aggregate → stats → plots

- Deterministic evaluation per seed
- Aggregated summaries across seeds
- Statistical comparisons per metric

Snakemake, Nix and Apptainer for reproducibility.

from concept to executable pipeline

METHODOLOGY

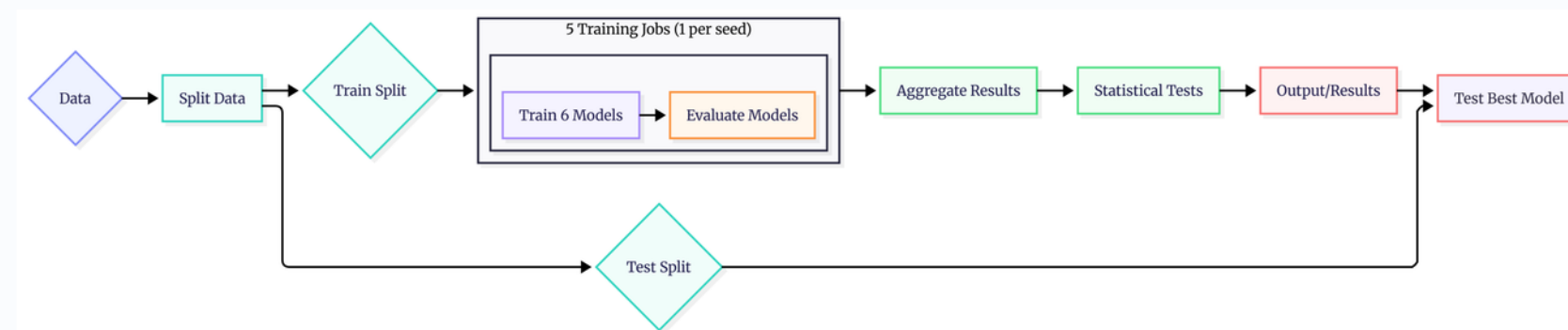
WHAT'S NEW AGAINST THE INTRO AND LITERATURE REVIEW

- Formalised train, eval and aggregate contracts (files and metadata)
- Time-aware split policy with holdout isolation
- Reproducibility stack: Nix, Snakemake and manifest hashes
- Explicit `treatment_id` schema for masked and unmasked variants

EXPERIMENTAL SCOPE

- 6 DRL algorithms × 5 seeds × 2 traces
- Deterministic evaluation per seed
- Baselines reported separately, descriptively

THE FULL SNAKEMAKE WORKFLOW



Dev split through to statistics; the holdout stays isolated.

CORE TESTS, PER METRIC

Shapiro–Wilk	Diagnostics
Friedman	Omnibus
Nemenyi	Pairwise
Kendall's W	Effect size

NEW ADDITIONS AGAINST THE INTRO AND LITERATURE REVIEW

- Wilcoxon non-parametric CI on paired differences
- Confidence curves (delta against p-value)
- Page trend test for ordered treatments
- CD diagram input for rank visualisations

All tests run per metric, with seeds as blocks.

CLUSTER EXECUTION

- Apptainer export of the Nix environment
- Slurm-backed Snakemake runs
- Full seed sweep on dev splits

BEST ALGORITHM EVALUATION

- Pareto selection over primaries and secondaries
- Tie-breakers: `avg_waiting` → `avg_slowdown` → `cpu_utilization`
- Holdout evaluation across all seeds

INTERPRETATION & REPORTING

- PNG plots and CSV tables for Typst
- Combined DRL against baseline visuals, descriptive
- Final discussion of algorithm choice and masking impact

STATISTICS

Carrasco, García, Rueda, Das & Herrera (2020). Recent trends in the use of statistical tests for comparing algorithms. *Swarm Evol. Comput.* 54, 100665. The framework followed here.

Demšar (2006). Statistical Comparisons of Classifiers over Multiple Data Sets. *JMLR* 7, 1–30. Nemenyi and the CD diagram.

Landis & Koch (1977). The Measurement of Observer Agreement for Categorical Data. *Biometrics* 33(1), 159–174. The interpretation used for Kendall's W.

Derrac et al. (2011). A practical tutorial on the use of nonparametric statistical tests. *Swarm Evol. Comput.* 1(1), 3–18. The Page trend test.

REPRODUCIBILITY

Mölder et al. (2025). Sustainable data analysis with Snakemake. *F1000Research* 10(33). The run pipeline.

Dolstra, de Jonge & Visser (2004). Nix: A Safe and Policy-Free System for Software Deployment. *USENIX LISA*. The pinned environment.